

Periodic Classification of Elements

I] Textbook Questions:

★ 1 Mark Objectives: ★

1. Choose the correct option and rewrite the statement.

- The number of electrons in the outermost shell of alkali metals is 1.
(i) 1 (ii) 2 (iii) 3 (iv) 7
- Alkaline earth metals have valency 2. This means that their position in the modern periodic table is in **Group 2**.
(i) Group 2 (ii) Group 16 (iii) Period 2 (iv) d-block
[2025]
- Molecular formula of the chloride of an element X is XCl . This compound is a solid having high melting point. Which of the following elements be present in the same group as X.
Ans : **(i) Na** (ii) Mg (iii) Al (iv) Si
- In which block of the modern periodic table are the nonmetals found?
(i) s-block Ans: **(ii) p-block** (iii) d-block (iv) f-block
[2023, 2024, 2026]

2. Rearrange the columns 2 and 3 to match with the column 1

[2024]

Ans:

Column 1	Column 2	Column 3
i. Triad	Average of the first and the third atomic mass	Dobereiner
ii. Octave	Properties of the eighth element similar to the first	Newlands
iii. Atomic number	Positive charge on the nucleus	Moseley
iv. Period	Sequential change in molecular formula	Mendeleev
v. Nucleus	Concentrated mass and positive charge	Rutherford
vi. Electron	Lightest and negatively charged particle in all the atoms	Thomson

3. Write the name and symbol of the element from the description.

- | | | |
|--|--------------------------|--------|
| a. The atom having the smallest size | Ans. Helium (He) | |
| b. The atom having the smallest atomic mass | Ans. Hydrogen (H) | [2025] |
| c. The most electronegative atom | Ans. Flourine (F) | [2025] |
| d. The noble gas with the smallest atomic radius | Ans. Helium (He) | [2025] |
| e. The most reactive non-metal | Ans. Flourine (F) | |

4. Name the following.

i. Write the names from the description.

- | | | |
|--|------------------------------------|--------|
| a. The period with electrons in shells K, L, and M | Ans. 3rd period | |
| b. The group with valency zero | Ans. Group 18 or zero group | |
| c. The family of non-metals having valency one | Ans. Halogens | |
| d. The family of metals having valency one | Ans. Alkali metals | |
| e. The family of metals having valency 2 | Ans. Alkaline earth metals | |
| f. Metalloids in the second and third periods | Ans. Boron and Silicon | [2026] |
| g. Non-metals in the third period | Ans. Phosphorus and Sulphur | [2026] |
| h. Two elements having valency 4 | Ans. Carbon and Silicon | [2026] |
| i. The unit in which atomic radius is expressed | Ans. Picometer | |

★ **2 Marks Questions:** ★

Q. Give Reasons:

- 1. Atomic radius goes on decreasing while going from left to right in a period.** [2025]

Ans:

1. Atomic radius is the distance between the centre of the atom and its outermost shell.
2. While going from left to right within a period, the atomic number increases one by one, meaning the positive charge on the nucleus increases by one unit at a time.
3. However, the additional electron gets added to the same outermost shell.
4. Due to the increased nuclear charge, the electrons are pulled towards the nucleus to a greater extent and thereby the size of the atom decreases.

- 2. Metallic character goes on decreasing while going from left to right in a period.** [2023]

Ans:

1. Metals tend to lose the valence electrons to form cations having a stable noble gas configuration.
2. This tendency of an element called electropositivity is the metallic character of that element. While going from left to right within a period the outermost shell remains the same.
3. However, the positive charge on the nucleus goes on increasing while the atomic radius goes on decreasing and thus the effective nuclear charge goes on increasing.
4. As a result of this the tendency of atom to lose valence electrons decreases within a period from left to right. Hence, Metallic character goes on decreasing while going from left to right in a period.

3. Atomic radius goes on increasing down a group.*[2022, 2026]***Ans:**

1. while going down a group the atomic size goes on increasing.
2. This is because while going down a group a new shell is added.
3. Therefore, the distance between the outermost electron and the nucleus goes on increasing.
4. As a result of this the atomic size increases despite the increased nuclear charge.

4. Elements belonging to the same group have the same valency.*[2020, 2021, 2023]***Ans:**

1. Valency is the number of electrons donated, accepted, or shared by an atom to complete the octet.
2. It depends on the number of electrons present in the outermost shell i.e. valence electrons of the atom
3. Elements in the same group have the same number of valence electrons. Example in Group 1, alkali metals the number of valence electrons is 1.
4. Hence, elements in the same group have the same valency.

5. The third period contains only eight elements even though the electron capacity of the third shell is 18.**Ans:**

1. The electrons holding capacity of the shells of an atom is given by the formula $2n^2$.
2. Based on this the number of electrons in the first orbit is 2, second is 8 and third is 18.
3. But the third period also contains only eight elements due to the law of electron octet.
4. It states that if the third shell is the last shell, then it will only contain a maximum of 8 electrons.

Q. Write short notes:**1. Position of isotopes in the Mendeleev's and the Modern periodic table.**

Ans:

1. Isotopes were discovered long after Mendeleev put forth the periodic table.
2. Mendeleev arranged the elements in an increasing order of atomic masses. As isotopes have the same chemical properties but different atomic masses, no fixed positions could be given to the isotopes in Mendeleev's table.
3. However, the modern periodic table is based on atomic numbers of elements.
4. As isotopes of an element have the same atomic number the isotopes could be placed in the same group and the radioactive isotopes were placed in two separate series below at the bottom of the periodic table.

2. Mendeleev's periodic law:**Ans:**

1. Mendeleev considered the fundamental property of an element namely the atomic mass as standard.
2. He arranged 63 elements known at that time in an increasing order of their atomic masses.
3. Mendeleev found that the elements with similar physical and chemical properties repeat after a definite interval.
4. Based on this finding, Mendeleev, stated the following periodic law "Properties of elements are periodic function of their atomic masses".

★ 3 Marks Questions: ★**Q.1 Answer the following.****i. An element has its electronic configuration as 2,8,2. Now answer the following questions.****a. What is the atomic number of this element?** [2020, 2025]**Ans:** The atomic number of this element is 12. This element is magnesium (Mg).**b. What is the group of this element?** [2020]**Ans:** The group of this element is 2.**c. To which period does this element belong?** [2020, 2020, 2025]**Ans:** This element belongs to third period.**d. With which of the following elements would this element resemble? (Atomic numbers are given in brackets) N(7), Be(4), Ar(18), Cl(17)****Ans:** Be (4)

Q. Write short notes:**2. Structure of the Modern Periodic table.****Ans:**

1. The Modern Periodic table contains seven horizontal rows called the periods 1 to 7.
2. Similarly, eighteen vertical columns are the groups 1 to 18.
3. Apart from these seven rows, two separate rows are placed at the bottom of the periodic table.
4. These are called Lanthanide series and Actinide series respectively.
5. There are 118 boxes (Elements) in the periodic table including the two series.
6. The entire periodic table is divided into four blocks, viz, s-block (Group 1 & 2), p-block (Groups 13 to 18), d-block (Groups 3 to 12) and f-block (Lanthanides and Actinides).

II] Extra - Previous Years' Questions: (PYQs)**★ 1 Mark Objectives: ★**

1. Write the electronic configuration of 'Aluminium'. [2025]

Ans: Aluminium has atomic number 13. Electronic configuration: **2, 8, 3**

2. Complete the correlation: Group 1 : Alkali metals :: **Group 17**: Halogens [2025]

★ 2 Marks Questions: ★

1. State Dobereiner's law of triad. Give one example of it. [2024]

Ans:

1. German scientist Dobereiner suggested that properties of elements are related to their atomic masses.
2. He made groups of three elements each, having similar chemical properties and called them triads.
3. He arranged the three elements in a triad in an increasing order of atomic mass and showed that the atomic mass of the middle element was approximately equal to the mean of the atomic masses of the other two elements.

Sr. No.	Triad	Element - 1 Actual atomic mass(a)	Element - 2		Element - 3 Actual atomic mass (c)
			Mean = $\frac{a+c}{2}$	Actual atomic mass	
1	Li, Na, K	Lithium (Li) 6.9	Sodium $\frac{6.9 + 39.1}{2} = 23.0$	(Na) 23.0	Potassium (K) 39.1

4.

2. Define Periods and Groups of modern periodic table.

[2023]

Ans:

1. The horizontal rows in the modern periodic table are called as Periods.
2. The vertical columns in the modern periodic table are called as Groups.

★ 3 Marks Questions: ★**1. Answer the following sub questions regarding the modern periodic table.**

[2025]

1. Write modern periodic law.**Ans:** Properties of elements are periodic function of their atomic numbers.**2. Write number of periods and groups.****Ans:** There are seven periods and 18 groups in the modern periodic table.**3. What is the valency of halogens?****Ans:** The valency of halogens is 1.**2. From the elements of second period of the modern periodic table given below, answer the following question:**

[2024]

³ Li	⁴ Be	⁵ B	⁶ C	⁷ N	⁸ O	⁹ F	¹⁰ Ne
Lithium	Beryllium	Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon
6.941	9.012	10.811	12.011	14.007	15.999	18.908	20.180

- a. Name the elements in which both the shells are completely filled with electrons.

Ans: Neon has both the shells completely filled with electrons i.e. 2, 8.

- b. Name the element which has same number of electrons in both the shells.

Ans: Beryllium has same number of electrons in both the shells i.e. 2, 2.

- c. Which is the most electropositive element in this period?

Ans: Lithium is the most electropositive element in this period.

3. Select the appropriate options and complete the following paragraph: [2024]

(metals, non-metals, metalloids, four, seven, s-block, p-block, d-block, f-block).

On the basis of electronic configuration, elements in the modern periodic table are classified into **four** blocks. Groups 1 and 2 elements are included in **s-block** and all these elements are metals (except Hydrogen). Group 13 to 18 elements are included in **p-block**. This block contains metals, non-metals and metalloids. Group 3 to 12 elements are included in **d-block** and all the elements are **metals** elements shown at the bottom of the periodic table i.e. Lanthanides and Actinides constitute **f-block** and all these elements are metals.

4. The electronic configuration of an element X is 2, 8, 8, 2. [2023]

a. What is the atomic number of this element?

Ans: The atomic number of this element is 20.

b. To which group does this element belong?

Ans: This element belongs to the 2nd Group.

c. In which period does this element lie?

Ans: This element lies in the 4th Period.

5. To which group does the halogen family belong? Write any four halogens. [2022]

Ans: Halogen family belongs to the group 17. Fluorine, chlorine, bromine, and iodine are the four halogens.

6. Write the demerits of the Mendeleev's Periodic table. [2022]

Ans:

1. The whole number atomic mass of the elements cobalt (Co) and nickel (Ni) is the same. Therefore, there was an ambiguity regarding their sequence in Mendeleev's periodic table.
2. Isotopes were discovered long time after Mendeleev put forth the periodic table.
3. As isotopes have the same chemical properties but different atomic masses, a challenge was posed in placing them in Mendeleev's periodic table.
4. When elements are arranged in an increasing order of atomic masses, the rise in atomic mass does not appear to be uniform.
5. It was not possible, therefore, to predict how many elements could be discovered between two heavy elements.
6. No fixed position could be given to hydrogen which resembles alkali metals and halogens in the periodic table.